Day 05 — Ethernet and MAC Addresses

Ethernet Switching and MAC Addressing

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Part 1: Theory Questions and Answers

1. What is Ethernet?

Ethernet is a widely used networking technology for connecting devices in a wired local area network (LAN). It defines how devices transmit and receive data over the network.

2. What is an Ethernet frame?

An Ethernet frame is the data unit transmitted over an Ethernet network. It contains fields such as the destination MAC address, source MAC address, type or length information, payload, and frame check sequence.

3. What is a MAC address?

A MAC address is a link-layer address used to identify a network interface on a local network. Ethernet MAC addresses are commonly displayed as six pairs of hexadecimal digits.

4. What does MAC stand for?

MAC stands for Media Access Control.

5. What is the difference between a MAC address and an IP address?

A MAC address is used for communication on a local network link, while an IP address provides logical addressing for communication across networks. MAC addresses operate at the Data Link Layer, whereas IP addresses operate at the Network Layer. An IP address can change when a device joins a different network.

6. What is a MAC Address Table?

A MAC Address Table is a table maintained by a network switch that records learned MAC addresses and the switch ports associated with them.

7. How does a switch learn MAC addresses?

A switch examines the source MAC address of each incoming Ethernet frame. It records that address alongside the port on which the frame arrived, allowing it to identify the port associated with that device.

8. What is a source MAC address?

The source MAC address identifies the network interface that sent an Ethernet frame on the local link.

9. What is a destination MAC address?

The destination MAC address identifies the intended recipient of an Ethernet frame on the local link. It can identify an individual device or represent a broadcast or multicast destination.

10. What is an unknown unicast frame?

An unknown unicast frame is a frame whose destination MAC address is not currently present in the switch's MAC Address Table for the relevant forwarding context.

11. What is flooding?

Flooding occurs when a switch sends a frame out multiple eligible ports rather than forwarding it to one known destination port. For an unknown unicast frame, the switch typically floods it to eligible ports in the same VLAN, excluding the incoming port.

12. What is known unicast forwarding?

Known unicast forwarding occurs when a switch finds the destination MAC address in its MAC Address Table and sends the frame through the associated port instead of flooding it to other ports.

13. At which OSI layer does Ethernet switching primarily operate?

Ethernet switching primarily operates at Layer 2, the Data Link Layer, of the OSI model.

14. Why does a switch maintain a MAC Address Table?

The table helps a switch forward frames efficiently to the correct ports. This reduces unnecessary traffic on other ports and improves local network communication.

15. Why is Ethernet knowledge important in cybersecurity?

Understanding Ethernet helps cybersecurity professionals interpret packet captures, identify devices on a local network, understand switch forwarding behavior, and investigate suspicious traffic. It also provides useful background for analyzing MAC address spoofing and other Layer 2 security issues.

Part 2: Screenshot Evidence

Windows Command Prompt output for ipconfig /all, getmac, and arp -a.

Screenshot 1 — ipconfig /all (Part 1)

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C:\Windows\System32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Anas_Bilal
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 10-7C-61-67-01-9F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

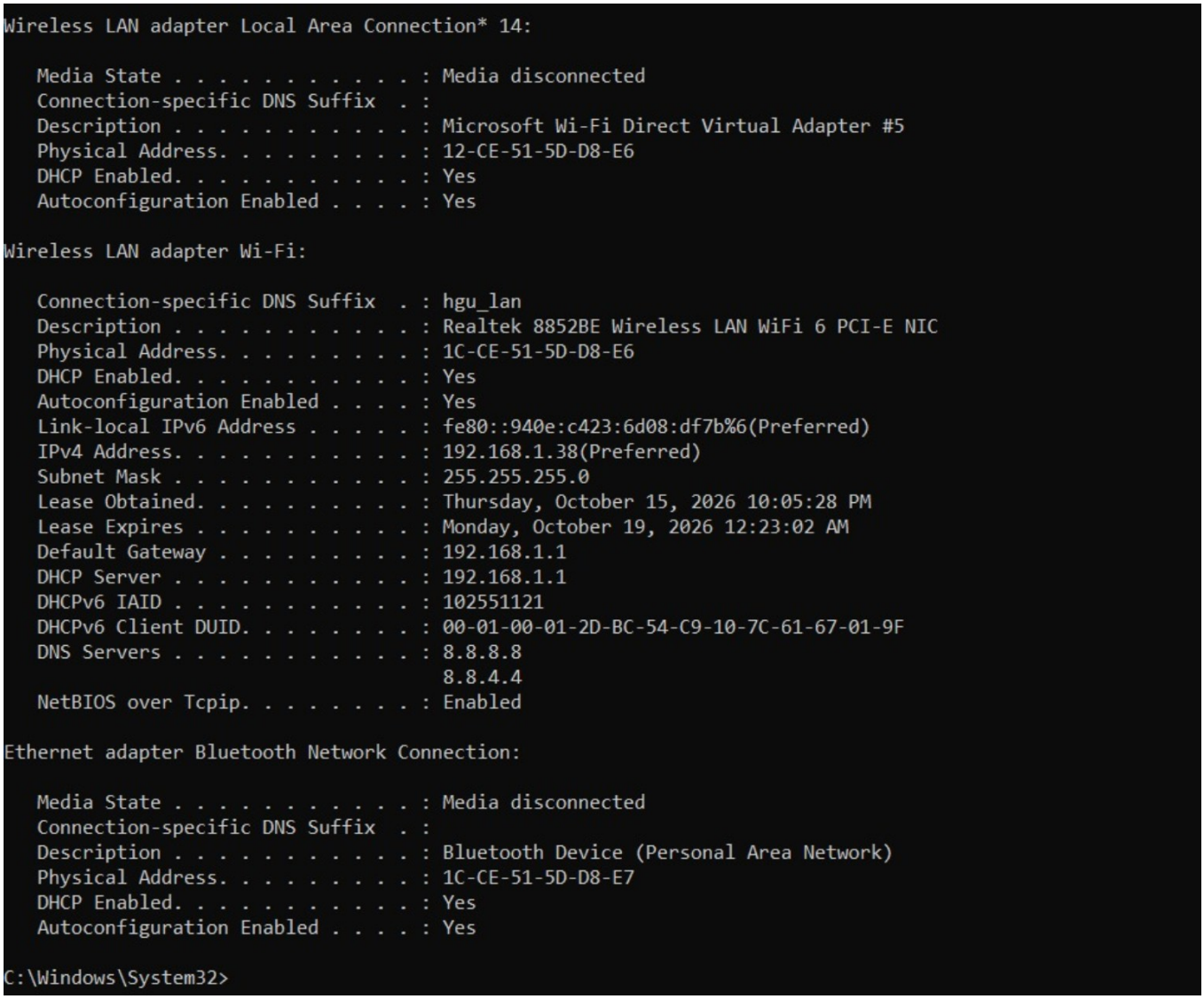
Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-0D
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::72b7:b0e9:6d91:a4%13(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 856293415
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-BC-54-C9-10-7C-61-67-01-9F
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 13:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #4
   Physical Address. . . . . . . . . : 1E-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

Screenshot 2 — ipconfig /all (Part 2)



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Wireless LAN adapter Local Area Connection* 14:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #5
   Physical Address. . . . . . . . . : 12-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

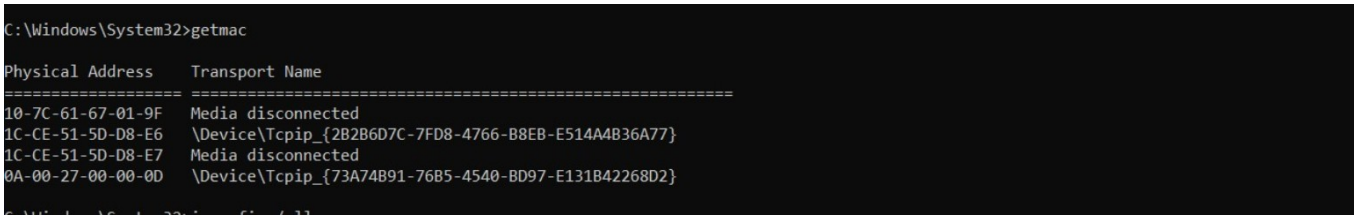
   Connection-specific DNS Suffix  . : hgu_lan
   Description . . . . . . . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
   Physical Address. . . . . . . . . : 1C-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::940e:c423:6d08:df7b%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.38(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, October 15, 2026 10:05:28 PM
   Lease Expires . . . . . . . . . . : Monday, October 19, 2026 12:23:02 AM
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 102551121
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-BC-54-C9-10-7C-61-67-01-9F
   DNS Servers . . . . . . . . . . . : 8.8.8.8
                                       8.8.4.4
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

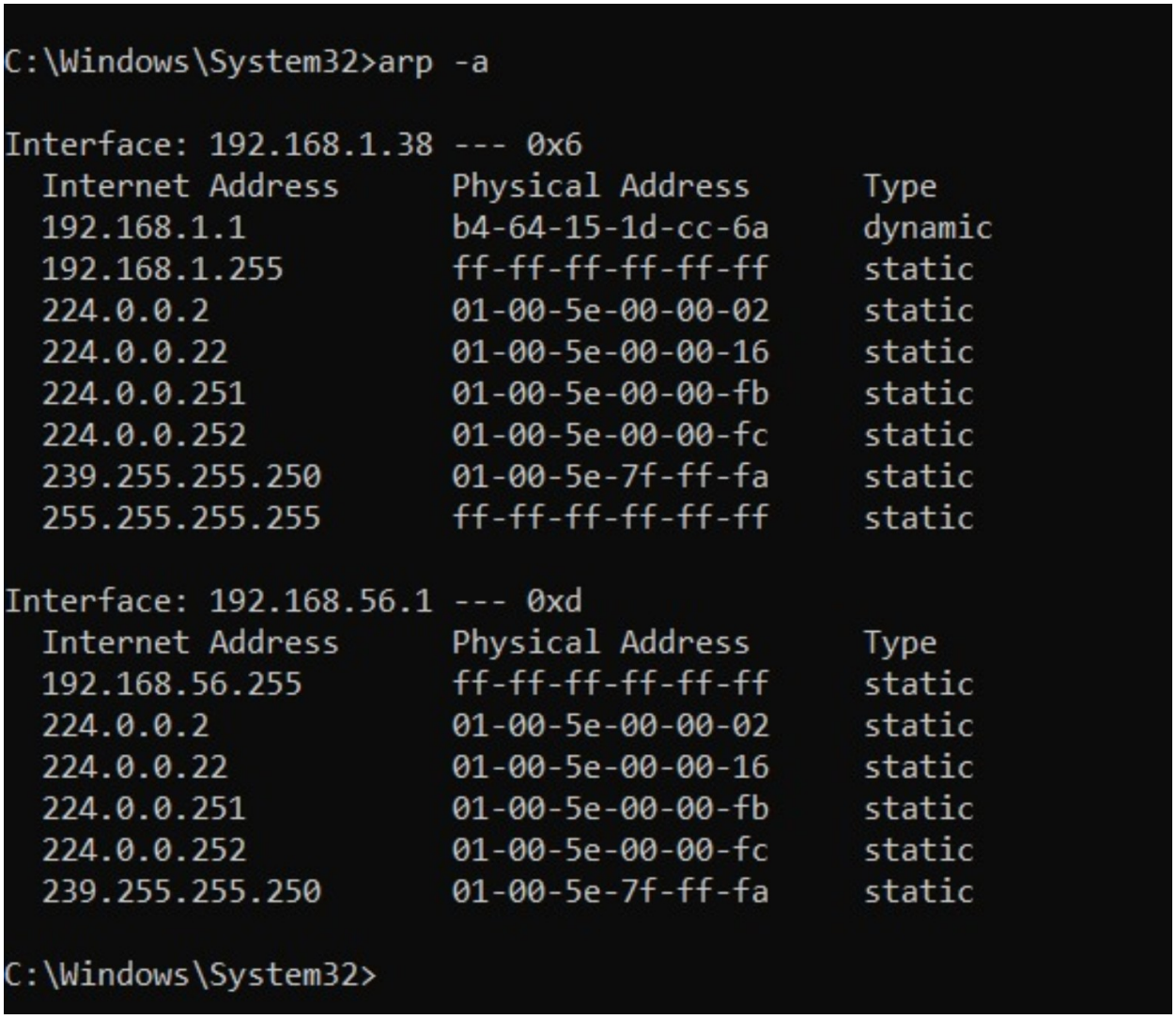
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 1C-CE-51-5D-D8-E7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Windows\System32>
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Screenshot 3 — getmac



Screenshot 4 — arp -a



Conclusion

Ethernet and MAC addressing are fundamental to local network communication. Understanding how switches learn addresses, maintain forwarding tables, and handle unknown destinations helps build a stronger foundation for network troubleshooting and cybersecurity analysis.